

Inferring Manifolds Using Gaussian Processes

From noisy local geometry to probabilistic reconstruction

Based on Dunson and Wu, *Biometrika* (2026)

Dimension reduction is not manifold reconstruction

Classical manifold learning

$$y_i \in \mathbb{R}^D \mapsto z_i \in \mathbb{R}^d$$

- coordinates for visualization or downstream learning
- manifold between observations: **not estimated**
- original data: **not denoised**

Manifold estimation

$$\{y_i\}_{i=1}^n \mapsto \{\hat{y}_i\}_{i=1}^n \in \widehat{\mathcal{M}} \subset \mathbb{R}^D$$

- fitted values in the ambient space
- interpolation between observations
- local structures: tangent space or curvature

Statistical target of MrGap

Observations fluctuate around an unknown embedded manifold:

$$y_i = \iota(x_i) + \eta_i, \quad x_i \sim P \text{ on } \mathcal{M}, \quad \eta_i \sim \mathcal{N}(0, \sigma^2 I_D).$$

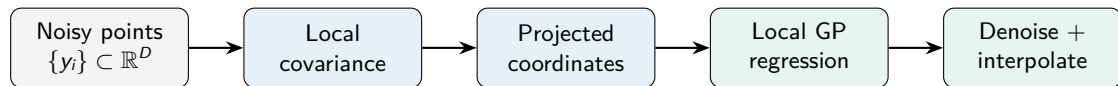
Given

- noisy points $y_i \in \mathbb{R}^D$
- intrinsic dimension $d \ll D$
- unknown sampling density P

Estimate

- denoised points $\hat{y}_i$
- local charts of $\iota(\mathcal{M})$
- new points on the fitted manifold

Global geometry becomes local regression



Central reduction

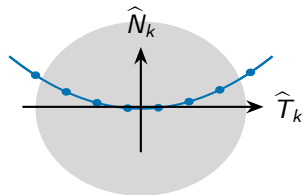
A global manifold problem becomes many low-dimensional, errors-in-variables regressions.

Local covariance

For a center y_k , retain neighbors inside radius ϵ :

$$C_{n,\epsilon}(y_k) = \frac{1}{n} \sum_{i=1}^n (y_i - y_k)(y_i - y_k)^\top \mathbf{1}\{\|y_i - y_k\| \leq \epsilon\}.$$

- first d eigenvectors: high local variation
- remaining $D - d$: normal directions
- no uniform-design assumption on P



Local coordinates from eigendirections

Write

$$C_{n,\epsilon}(y_k) = U_{n,\epsilon}(y_k) \Lambda_{n,\epsilon}(y_k) U_{n,\epsilon}(y_k)^\top.$$

The rotation $U_{n,\epsilon}(y_k)^\top$ defines two projections:

$$\underbrace{P_{y_k}(y)}_{\mathbb{R}^D \rightarrow \mathbb{R}^d} = J^\top U_{n,\epsilon}(y_k)^\top (y - y_k), \quad \underbrace{P_{y_k}^\perp(y)}_{\mathbb{R}^D \rightarrow \mathbb{R}^{D-d}} = \bar{J}^\top U_{n,\epsilon}(y_k)^\top (y - y_k).$$

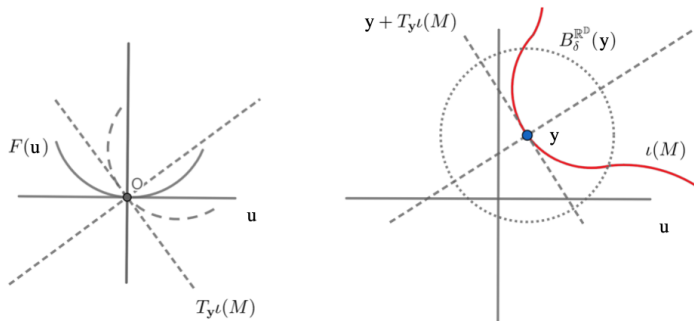
Roles

$P_{y_k}(y)$: GP predictor; $P_{y_k}^\perp(y)$: multivariate GP response.

Why a local graph is enough

Near a clean point $y \in \iota(\mathcal{M})$, rotate and translate the manifold:

$$\Phi(u) = y + U \begin{bmatrix} u \\ F(u) \end{bmatrix}, \quad F: \mathbb{R}^d \supset O \rightarrow \mathbb{R}^{D-d}.$$



Source: Dunson and Wu (2026), Figure 2.

The key geometric relaxation

Classical local methods try to estimate the true tangent space $T_{\iota(x_k)}\iota(\mathcal{M})$.

Stronger requirement

$$\hat{T}_k \approx T_{\iota(x_k)}\iota(\mathcal{M})$$

Accurate tangent recovery at every sample.

MrGap requirement

$P_{y_k}|_{\iota(\mathcal{M})}$ is locally invertible

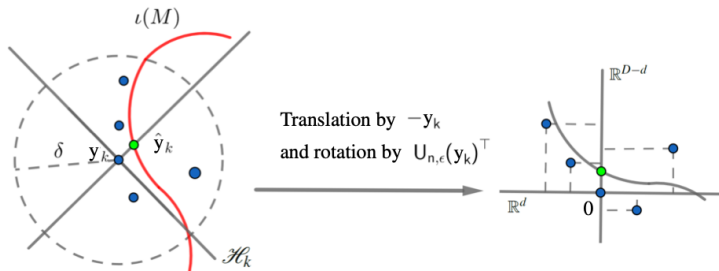
Enough to write the manifold as a graph over $H_k = y_k + \hat{T}_k$.

Useful coordinates do not require a perfect tangent estimate.

Neighbors become regression pairs

For $y_{k,j} \in B_\delta(y_k) \setminus \{y_k\}$, construct

$$w_{k,j} = P_{y_k}(y_{k,j}), \quad z_{k,j} = P_{y_k}^\perp(y_{k,j}).$$



Source: Dunson and Wu (2026), Figure 3.

Local Gaussian process

The projected pairs obey an errors-in-variables representation:

$$w_i = u_i + \eta_{i,1}, \quad z_i = F(u_i) + \eta_{i,2}.$$

Independently for $r = 1, \dots, D - d$,

$$f_r \sim \text{GP}(0, C), \quad C(u, u') = A \exp \left\{ -\frac{\|u - u'\|^2}{\rho} \right\}.$$

- shared A, ρ, σ across local fits
- empirical Bayes via summed marginal likelihood
- prediction at $u = 0$: denoising
- prediction at new u : interpolation

Where should A , ρ , and σ come from?

MrGap shares one hyperparameter triple across all local regressions in round t :

$$(\hat{A}^{(t)}, \hat{\rho}^{(t)}, \hat{\sigma}^{(t)}) = \arg \max_{A, \rho, \sigma} \sum_{k=1}^n \log p(Z_k \mid A, \rho, \sigma).$$

Layer	What is available
paper	summed marginal-likelihood objective; re-estimate each round
public <code>Mfit1.m</code>	denoising conditional on supplied A, ρ, sig ; no optimizer

Hyperparameters fixed in the current iteration scan

Manifold	round 1 (A, ρ, σ^2)	rounds 2–5 (A, ρ, σ^2)	provenance
Cassini	(.014, .2, .002)	(.048, .3, 2×10^{-5})	paper §4.1
$\mathbb{RP}^3$	(.05, 1, .0015)	(.08, 1.2, .0004)	reproduction setting
Torus	(.06, .2, .03)	(.2, 1.1, .007)	paper §4.2
Half-torus	(1.3, 5, .002)	(1.3, 5, .002)	reproduction setting

Why is the algorithm built from local pieces?

Geometric obstacle	Design choice	What it buys
no single global chart	overlapping patches	topology and changing axes
unknown local orientation	covariance eigenspaces	tangent/normal coordinates
manifold is curved	graph $F: \mathbb{R}^d \rightarrow \mathbb{R}^{D-d}$	intrinsic-dimension regression
observations are noisy	GP posterior mean	smooth correction and uncertainty
frames improve after cleaning	repeat all local steps	updated neighbors, frames, fits

Each component removes one obstruction to turning an unknown manifold into a regression problem.

Why predict only the normal displacement?

In an estimated local frame, a neighbor decomposes as

$$y_j - y_k = \hat{T}_k w_{k,j} + \hat{N}_k z_{k,j}, \quad z_{k,j} \approx F_k(w_{k,j}) + \text{noise}.$$

The manifold point above the chart origin is therefore estimated by

$$\hat{y}_k = y_k + \hat{N}_k \hat{F}_k(0).$$

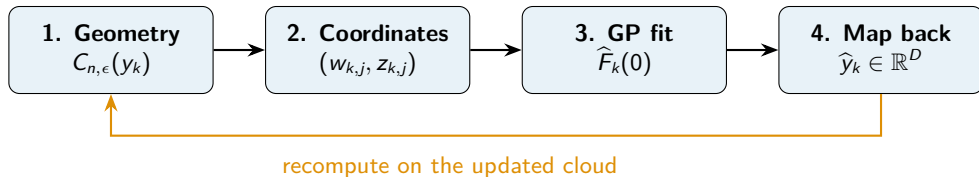
Keep the tangential coordinate fixed

Tangential motion changes location along the manifold; it does not identify the off-manifold error.

Learn the normal graph

GP averaging estimates curvature while shrinking noisy normal residuals; posterior covariance quantifies local uncertainty.

Denoising phase



Inputs and output

Inputs: $\{y_i\}, d, \epsilon, \delta$. Output: $\{\hat{y}_i\}$ plus local predictive covariances.

Prediction in the normal affine space

Partition the GP covariance over training predictors and 0:

$$\Sigma_k = \begin{bmatrix} \Sigma_{k,1} & \Sigma_{k,2} \\ \Sigma_{k,3} & \Sigma_{k,4} \end{bmatrix}.$$

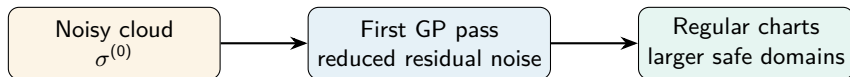
The local posterior mean maps back as

$$\hat{y}_k = y_k + U_{n,\epsilon}(y_k) \begin{bmatrix} 0 \\ \{\Sigma_{k,3}(\Sigma_{k,1} + \sigma^2 I)^{-1} Z_k\}^\top \end{bmatrix}.$$

Geometric interpretation

Tangential coordinate fixed at 0; correction occurs in the estimated normal affine space $A_k = y_k + \hat{N}_k$.

Why iterate the denoising map?



- the first correction changes distances and neighborhood membership
- reduced normal noise can sharpen the next local PCA frame
- the next GP is fitted in updated coordinates, not merely reapplied

Stopping is part of the estimator

Paper: stop when $|\hat{\sigma}^{(t)} - \hat{\sigma}^{(t-1)}|$ is small. Current fixed-parameter scan: this criterion is not evaluated.

Interpolation has two extra challenges

Given a local chart, a new point would be

$$\tilde{y}_{k,j} = y_k + U_{n,\epsilon}(y_k) \begin{bmatrix} \tilde{u}_{k,j} \\ \hat{F}_k(\tilde{u}_{k,j}) \end{bmatrix}.$$

Unknown domain

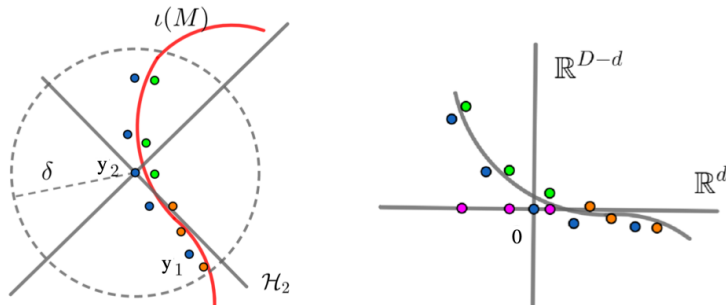
$\tilde{u}_{k,j}$ must remain inside the unknown chart domain O_k .

Chart seams

Independent predictions from overlapping charts can leave visible gaps.

Sequential interpolation stitches local charts

When processing chart k , reuse earlier interpolations that fall inside $B_\delta(y_k)$ as additional regression pairs.



Source: Dunson and Wu (2026), Figure 4.

Overlap information couples otherwise independent local GPs.

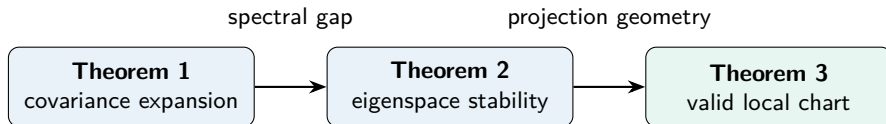
Choosing the geometric scales

Parameter	Geometric role	Statistical trade-off
ϵ	covariance neighborhood	eigenspace variance vs. curvature bias
δ	regression neighborhood	sample size vs. chart validity
ρ	GP correlation length	local flexibility vs. smoothness
σ	noise scale	shrinkage toward the fitted graph

Paper's strategy

Grid search over $\epsilon \leq \delta$; optimize A, ρ, σ by the summed local marginal likelihood.

Theory roadmap



Operational conclusion

With compatible $n, \epsilon, \delta, \sigma$, the projected data genuinely support a local regression representation of the manifold.

Theorem 1: covariance separates directions

After centering and rotating so that the first d axes are tangent,

$$C_{n,\epsilon}(y_k) = \epsilon^{d+2} \frac{|S^{d-1}|P(x_k)}{d(d+2)} \begin{bmatrix} I_d & 0 \\ 0 & 0 \end{bmatrix} + E_{n,\epsilon,\sigma}.$$

- signal occupies a rank- d block
- density $P(x_k)$ changes scale, not directions
- curvature, finite n , noise enter through E
- small E yields a local spectral gap

Implication

Local PCA remains informative under nonuniform sampling and Gaussian noise.

Theorem 2: eigenspaces are stable enough

Davis–Kahan perturbation control gives

$$U_{n,\epsilon}(y_k) = \begin{bmatrix} X_1 & 0 \\ 0 & X_2 \end{bmatrix} + O\left(\epsilon^{2\min(\beta-1,1)}\right),$$

where $X_1 \in O(d)$, $X_2 \in O(D-d)$.

What is identified

Tangent and normal *subspaces*, up to rotations within each.

What the algorithm needs

A coordinate projection that preserves local one-to-one geometry.

Theorem 3: projection really defines a chart

For $\epsilon^\beta < \delta < \tau_{\iota(\mathcal{M})}/16$, with high probability:

- ① noisy neighbors in $B_\delta(y_k)$ originate from the larger clean neighborhood;
- ② P_{y_k} is a diffeomorphism on $B_{3\delta}(y_k) \cap \iota(\mathcal{M})$;
- ③ the image $O_k \subset \mathbb{R}^d$ contains 0 and a nontrivial round ball.

Why this is the pivotal result

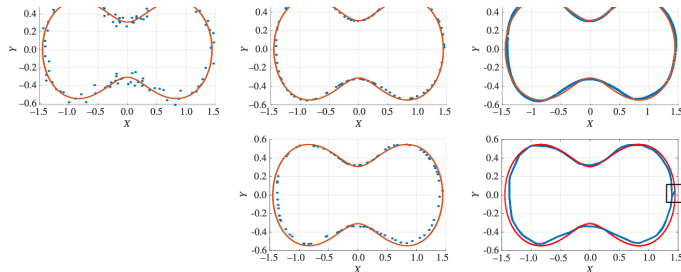
It justifies $F_k = P_{y_k}^\perp \circ P_{y_k}^{-1}$, turning manifold reconstruction into regression without observing tangent spaces.

Reading the assumptions statistically

Assumption	Why it appears	Practical warning
smooth closed $\mathcal{M}$	uniform local charts	boundaries need modification
positive smooth P	enough neighbors everywhere	sparse regions unstable
isotropic Gaussian noise	rotation preserves noise law	anisotropy can bias coordinates
small ϵ vs. reach	avoid chart folding	high curvature narrows locality
large n vs. ϵ^{-d}	covariance concentration	intrinsic dimension matters

The theory validates the reduction; it is not a full contraction-rate result for the GP estimator.

Evidence in one dimension: Cassini oval

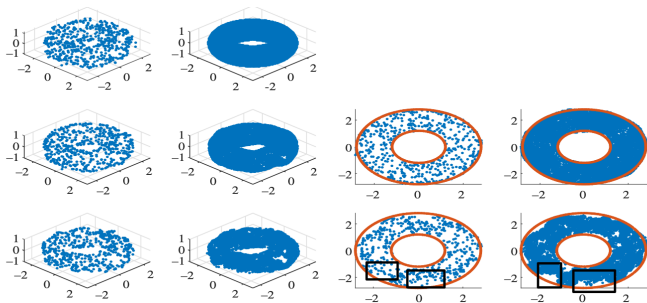


Output	G
noisy	0.0590
MrGap denoised	0.0224
MrGap interp.	0.0216
principal graph	0.0322

Principal graph: visible nonsmooth seam.

Source: Dunson and Wu (2026), Figure 6.

Evidence in two dimensions: torus



Output	G
noisy	0.1238
MrGap denoised	0.0544
MrGap interp.	0.0605
comparator den.	0.2196
comparator interp.	0.2424

MrGap preserves the annular geometry; the comparator shrinks the torus.

Source: Dunson and Wu (2026), Figure 7.

Reproducing and benchmarks

Component	Pilot design
Manifolds	Cassini oval, $\mathbb{RP}^3$, torus, half-torus
Ambient noise	$\sigma \in \{0, .01, .02, .04, .06, .08, .12\}$
Sample size	$n \in \{50, 100, 250, 500, 1000, 5000\}$; fresh samples at every n
Methods	MrGap after rounds 1–5; Manifold Fitting; oracle-tangent MrGap
Diagnostics	geometric RMSE G , paired reconstruction RMSE, runtime, median local count m_k

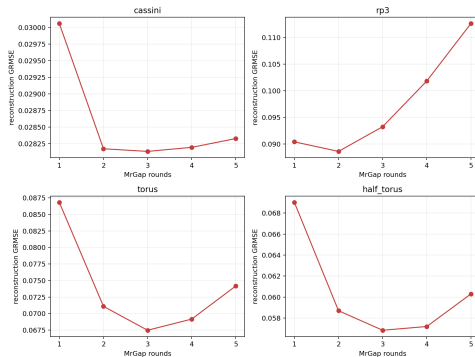
Pilot defaults

Manifold	n	σ	G		runtime (s)	
			MrGap-2	MF	MrGap-2	MF
Cassini	102	.04	.0282	.0285	.023	.012
$\mathbb{RP}^3$	500	.04	.0886	.0797	.129	.056
Torus	500	.12	.0711	.0691	.124	.056
Half-torus	400	.12	.0587	.0670	.100	.045

Accuracy: ranking changes with geometry; neither method dominates.

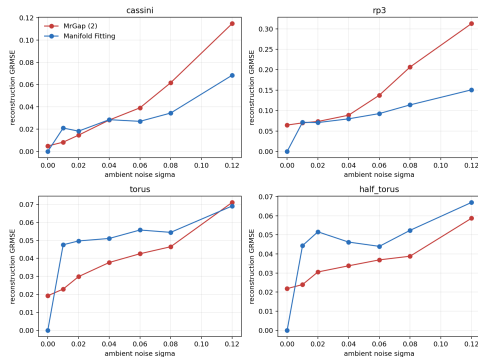
Cost: Manifold Fitting is about twice as fast at these small pilot sizes.

Useful MrGap stopping points are two or three rounds



Fixed-parameter diagnostic: round 1 uses the first tuple; rounds 2–5 reuse the final tuple. This is not an MLE convergence test.

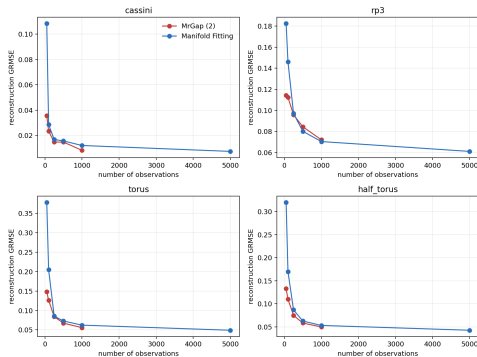
Noise sensitivity: failure is geometry-dependent



The winner can change across the same noise grid, so a single Cassini setting is not representative.

MF uses the known noise level

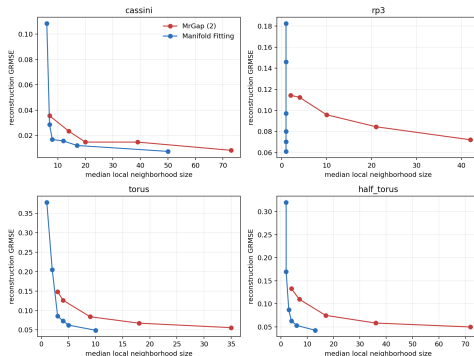
Global sample size: more data help only through local support



Each n is generated independently rather than subsampled from a fixed 100,000-point cloud. Curves are nonmonotone when fixed neighborhoods trade variance for curvature bias.

Exact MrGap pilot runs stop at $n = 1000$; $n = 5000$ is omitted by the resource guard.

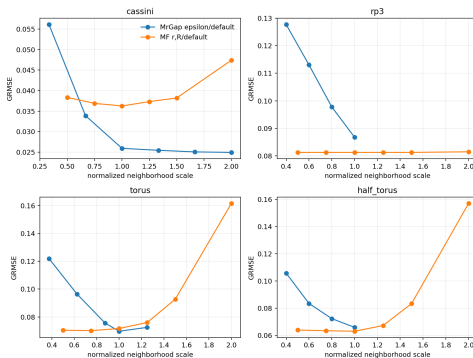
Effective sample size is the local count m_k



Global n is an incomplete predictor of error. On $\mathbb{RP}^3$, Manifold Fitting frequently reaches its official five-nearest-neighbor fallback because the cylindrical neighborhood is sparse.

Horizontal axis: median neighborhood size over reconstructed points.

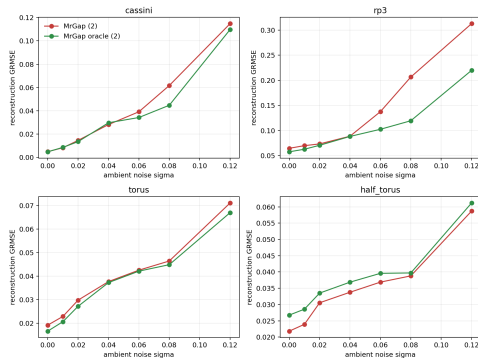
Bandwidth scan separates variance from curvature bias



Too-small neighborhoods provide unstable local fits; too-large neighborhoods average across curvature. The pilot optima differ substantially by manifold.

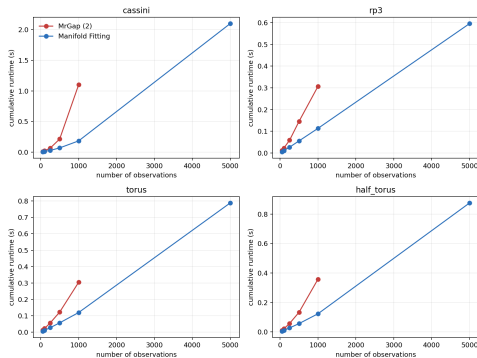
MF horizontal axis rescales both official radii; MrGap horizontal axis is ϵ .

Oracle tangents isolate the chart-estimation bottleneck



Oracle changes only the tangent basis; all other MrGap settings are fixed.

Runtime scaling: the current implementations target different regimes



Manifold Fitting is faster throughout the pilot.

Why couple neighboring local GPs?

Independent charts describe the *same physical manifold* on overlaps:

$$\Phi_k(u) = y_k + T_k u + N_k F_k(u) = y_\ell + T_\ell v + N_\ell F_\ell(v) = \Phi_\ell(v).$$

Independent MrGap fits

$$F_k \perp\!\!\!\perp F_\ell$$

Agreement encouraged only through sequentially reused interpolations.

Proposed coupled model

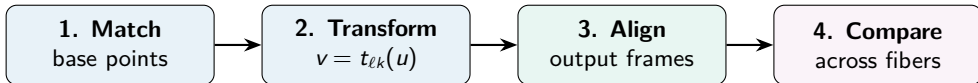
$$\text{Cov}\{F_k(u), F_\ell(v)\} \neq 0$$

Share information in sparse regions; reduce seams across charts.

The meaningful constraint is ambient agreement of Φ_k and Φ_ℓ , not equality of raw GP coordinates.

Why a naive cross-chart covariance is ill-defined

On $U_k \cap U_\ell$, four identifications are required:

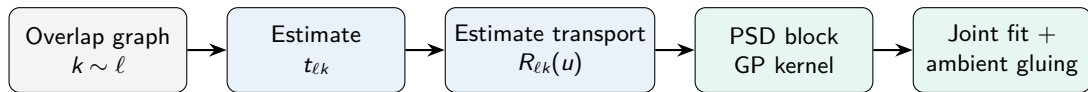


Normal-bundle caveat

$N_k F_k(u) \in \text{span}(N_k)$, a fixed transverse space; generally $N_k F_k(u) \notin N_{\Phi_k(u)} \mathcal{M}$.

Frame alignment alone is insufficient: both the input transition and output identification vary over the overlap.

A defensible coupled-GP construction



A schematic cross-covariance is

$$K_{k\ell}(u, v) = \kappa\{\text{dist}_{\text{graph}}(u, v)\} R_{k\ell}(u, v), \quad v \approx t_{\ell k}(u).$$

- connection-Laplacian methods: data-driven frame transport
- gauge-independent projected kernels: valid covariance under frame changes
- alternate geometry estimation and joint GP fitting

Coupling does not remove measurement-error bias

The local regression is errors-in-variables:

$$W = U + \xi, \quad Z = F(U) + \zeta.$$

A standard GP fitted to (W, Z) targets

$$m_{\sigma}(w) = \mathbb{E}(Z \mid W = w) = \frac{\{Fp\} * \varphi_{\sigma}(w)}{p * \varphi_{\sigma}(w)},$$

not $F(w)$ when predictor noise is nonnegligible.

Required error decomposition

Chart/frame estimation + measurement-error bias + GP estimation + cross-chart gluing error.

- frames and neighborhoods estimated from the same noisy data
- conditional GP variance omits coordinate and selection uncertainty

Discussion: a staged research program

Stage	Question	Evidence needed
1. Benchmark	Does coupling improve reconstruction?	independent GP vs. coupled GP vs. local polynomial/MMLS
2. Decouple	What is gained beyond shared data?	sample-split frames; fixed-chart experiments
3. Correct EIV	Does the model target F ?	latent- U likelihood or small-noise bias expansion
4. Global theory	Is the manifold consistently recovered?	Hausdorff/reach bounds and overlap control
5. Uncertainty	Are credible tubes calibrated?	simultaneous coverage including geometry estimation

References



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Backup: exact scale relations

For any $\beta > 1$, the main results assume

$$\sigma \leq \min \left\{ \frac{1}{\sqrt{-4(d+5)\log \epsilon}}, \frac{\epsilon^\beta}{\sqrt{12\log(2n)}} \right\}, \quad \epsilon^{-d-2\min(\beta-1,1)} \leq \frac{n}{\log n}.$$

- noise must be smaller than the covariance neighborhood
- effective local sample size must dominate perturbation terms
- δ additionally satisfies $\epsilon^\beta < \delta < \tau_{\ell}(\mathcal{M})/16$

Backup: covariance perturbation anatomy

Up to the leading factor ϵ^{d+2} , the error has block structure

$$E = \begin{bmatrix} \text{tangent bias + variance} & \text{tangent-normal coupling} \\ \text{tangent-normal coupling} & \text{normal leakage} \end{bmatrix}.$$

Bias sources

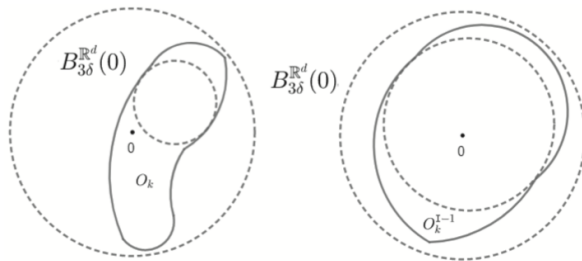
curvature, density derivatives, neighborhood truncation

Variance sources

finite local sample size and Gaussian perturbations

Davis–Kahan compares these terms with the leading eigengap.

Backup: iteration enlarges the safe interpolation region



Source: Dunson and Wu (2026), Figure 5.

- every chart image remains inside $B_{3\delta}^d(0)$
- decreasing fitted noise makes the guaranteed inner ball larger
- samples drawn from an estimated inner ball are less likely to leave the chart

Backup: geometric RMSE

For a finite set $Y = \{y_i\}_{i=1}^n$ and target $S \subset \mathbb{R}^D$,

$$G(Y, S) = \left\{ \frac{1}{n} \sum_{i=1}^n \text{dist}(y_i, S)^2 \right\}^{1/2}.$$

If Y_{true} densely samples $\iota(\mathcal{M})$, Theorem 4 bounds the discrepancy between $G(Y, Y_{\text{true}})$ and $G\{Y, \iota(\mathcal{M})\}$.

Evaluation caveat

$G(Y, \iota(\mathcal{M}))$ measures how far estimates lie from the manifold; it does not alone measure whether the entire manifold is covered.

Backup: algorithmic complexity and open questions

Component	Main cost driver	Possible improvement
neighborhoods	pairwise search in $\mathbb{R}^D$	approximate nearest neighbors
local covariance	$D \times D$ eigensystems	randomized/truncated SVD
local GPs	repeated kernel solves	inducing points or shared factors
interpolation	sequential chart coupling	parallel overlap graph

Open theory: GP contraction on estimated charts, anisotropic noise, boundaries, and globally calibrated uncertainty.